# ROWAN BRAD QUNI-GUDZINAS

## Autonomous AI systems researcher

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Earlier work is published under **Brad Gudzinas**.

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## SUMMARY

Systems researcher studying autonomous AI systems: how they are structured, how they behave in operation, how they fail, and how people can audit and govern them. Twenty years of systems work behind it: transportation network and GIS work from my first role in 2004, transportation network models at DVRPC and IBI Group, the national Tour-Based Model System for long-distance travel at the U.S. Federal Highway Administration, data-driven decision support at AARP, and machine-learning engagements at Deloitte. Since 2024 I have designed and run QNFO, an autonomous research system of 44 live agents, and published what it delivers, what it costs and where it fails. Open to autonomous and AI systems research, AI evaluation and assurance, AI governance and oversight research, and research program leadership for AI.

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## EXPERIENCE

### Founder and Systems Researcher

**QNFO** (independent research on autonomous AI systems) | 2024 to present

- Designed and operate an autonomous research system on Cloudflare (44 live workers) that drafts, referees, revises, versions, publishes and corrects papers, with standing objectives, health checks and a failure ledger.
- Reported how the system behaves in operation: Operating the Quniverse Fleet sets out its objectives, successes, failures and roadmap.
- Designed oversight methods for AI-assisted work: The Universal Ignorance Audit and Epistemic Legibility in AI-Assisted Science.
- Proposed JPCUB, a whole-system cost measure for computing (joules per correct answer), with a measurement protocol and an extension to stochastic and agentic LLM inference.
- Publish an open library of preprints with DOIs (papers.qnfo.org) and a knowledge graph that other tools can query by API.

### Executive Director and Founder

**Empowering Change** (nonprofit) | San Francisco, CA | 2023 to 2024

- Took the nonprofit from incorporation to public launch.
- Led an LLM legal-navigation tool for self-represented litigants: document drafting, procedural guidance and plain-language translation of legal terms.

### Platform Data Product Manager

**iManage** | 2022 to 2023

### Senior Product Manager

**Epsilon (Publicis Groupe)** | 2021 to 2022

- Managed data products built on predictive machine-learning models.

### Product Manager and Senior Methods Advisor

**AARP Public Policy Institute** | Washington, DC | 2016 to 2021

- Led the AARP Livability Index, a national decision-support system: 50+ data sources across 7 domains, scoring U.S. neighborhoods nationwide, across multiple public releases.
- Co-authored the Livability Index 2018 report and AARP policy work on housing affordability, including the 2016 analysis of Low-Income Housing Tax Credits.
- The Index and its method are cited in 20+ academic and policy studies.

### Senior Data Scientist

**The Advisory Board Company** | 2016

### Senior Associate, Data Analytics

**Deloitte Consulting LLP** | Washington, DC | 2015 to 2016

- Delivered analytics and machine-learning engagements for federal and public-sector clients.
- Co-authored a 2016 Transportation Research Board paper on updating the National Household Travel Survey with American Community Survey data.

### Data Analyst and Research Manager

**U.S. Department of Transportation, Federal Highway Administration** | Washington, DC | 2011 to 2015

- Specified the national Tour-Based Model System, a simulation of long-distance passenger travel across the United States, and directed the contractor team that built it.
- Co-directed the $10M U.S. DOT National Household Travel Survey.
- Managed a $1.5M+ federal research portfolio as a certified Contracting Officer's Representative: technical oversight, budgets, milestones and acceptance of contractor deliverables.
- Presented national long-distance travel data needs at the 2011 TRB National Household Travel Survey workshop.

### Earlier roles: transportation networks, systems planning and GIS

| Role | Organization | Period |
|---|---|---|
| Consultant: built and ran transportation network models | IBI Group | 2008 to 2011 |
| Systems Planner: built and ran transportation network models | Delaware Valley Regional Planning Commission | 2005 to 2008 |
| GIS Analyst | Cross County Connection TMA | 2004 to 2005 |

## SELECTED WORK

1. Operating the Quniverse Fleet: Objectives, Successes, Failures, and Roadmap of an Autonomous Cloud Research System (2026)
2. The Universal Ignorance Audit (2026)
3. Epistemic Legibility in AI-Assisted Science (2026)
4. Joules-per-Solution for Stochastic and Agentic Inference (2026)
5. Tour-Based National Model System, FHWA (2014, as Brad Gudzinas)
6. AARP Livability Index 2018 report (co-author, as Brad Gudzinas)

Every QNFO paper is prepared with an AI-assisted research pipeline; the author is responsible for its content. Full library: papers.qnfo.org. All records: Zenodo.

## EDUCATION

- **M.S., City and Regional Studies**, Rutgers University, Edward J. Bloustein School of Planning and Public Policy (2008)

- **B.A., Urban Studies and Community Planning**, Rutgers University-Camden (2004)

## CERTIFICATIONS

- Project Management Professional (PMP), Project Management Institute (2016)
- Contracting Officer's Representative (COR), Defense Acquisition University (2011)
- American Institute of Certified Planners (AICP), American Planning Association (2009)

## SKILLS

Autonomous and agentic system design · operating and monitoring multi-agent systems · failure analysis · oversight and audit methods for AI-assisted work · systems modeling · travel-demand and simulation modeling · statistical modeling and applied machine learning · data-driven decision support · GIS · research program management · federal contract oversight

Tools I direct and work with: Python, SQL, Cloudflare Workers, GIS (ArcGIS, QGIS), LLM agent frameworks.